

DATABASE MANAGEMENT SYSTEMS (CS301)

ACID Properties Cheat Sheet

Learning Objectives

Understand database transactions, ACID properties, transaction states, and their importance in maintaining data consistency and reliability.

Introduction to Transactions

A transaction is a sequence of database operations executed as a single logical unit of work. A transaction must either complete entirely or not execute at all.

What are ACID Properties?

ACID properties ensure reliable processing of database transactions and help maintain data integrity even during failures.

Atomicity

Atomicity means 'all or nothing'. If any part of a transaction fails, the entire transaction is rolled back.

Example of Atomicity

During a bank transfer, money must be deducted from one account and credited to another. If one operation fails, both operations are cancelled.

Consistency

Consistency ensures that a transaction takes the database from one valid state to another valid state while preserving integrity constraints.

Isolation

Isolation ensures that concurrent transactions do not interfere with each other. Each transaction behaves as if it is executing alone.

Durability

Durability guarantees that once a transaction is committed, its changes are permanently stored even if a system crash occurs.

Transaction States

- Active State
- Partially Committed State
- Committed State
- Failed State
- Aborted State

ACID Summary Table

Atomicity → All or Nothing
Consistency → Valid State Preservation
Isolation → Independent Execution
Durability → Permanent Storage

Importance of ACID Properties

- Prevents data corruption
- Ensures transaction reliability
- Maintains database consistency
- Supports concurrent users safely

Applications

- Banking Systems
- E-Commerce Platforms
- Airline Reservation Systems
- Hospital Management Systems

Exam Important Questions

1. Define a transaction.
2. Explain ACID properties.
3. What is Atomicity?
4. Explain transaction states.
5. Discuss the importance of ACID properties.

Summary

ACID properties are fundamental concepts in transaction management. They ensure that database transactions are reliable, consistent, and fault-tolerant.